



The
Best
Open
Source
Cloud
Computing
Simulators

Cloud computing is here to stay because of the great advantages that it offers. Service providers offering cloud computing need to evaluate their services from time to time. Real-time and real world evaluation can prove to be costly and impractical, so simulation offers an easy way out. This article introduces a selection of open source cloud computing simulators.

The cloud and cloud computing are among the more recent advances in the field of information technology. IT infrastructure and applications are provided to end users as services, using a standard pay-as-per-use model. Cloud computing has become widely prevalent because of recent advances in areas like virtualisation, grid computing, Web computing, utility computing and related technologies. Cloud computing is a powerful tool for accessing high performance computing (HPC) and storage infrastructure at reduced prices via the Internet. Some of the key characteristics of cloud computing are: on-demand access, elasticity, cost-effectiveness, scalability, minimal infrastructure management and location independence. Trillions of dollars are being invested in cloud computing today.

It is not possible, in the real world, for systems administrators, cloud specialists and even researchers to have actual cloud infrastructure to perform real-time experiments and implement new algorithms and methodologies. Before real-time implementation, it is

essential to first measure performance and take serious note of all possible security issues. To address such challenges and issues, modelling and simulation technologies come to our rescue. The need for a cloud computing simulator arises in order to witness an implementation scenario in real-time. Cloud simulators play a crucial role in reducing the complexity of the infrastructure, in executing new algorithms, analysing security threats and measuring the overall quality and performance of the infrastructure.

Several cloud computing simulators are being developed to assist researchers, systems administrators, cloud specialists and network administrators in measuring the real-time performance of cloud computing environments. Getting the right tool for a given scenario or knowing what features each tool has is a challenging and complex task.

Some simulators are free and open source, while others are commercial. As free and open source simulators provide an environment for deep learning and experimentation, the trend is to specifically work on these simulators for all types of complex and real world problems of cloud computing.